

Adeola Damilola Kola-Olaleye

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EDUCATION

Nile University of Nigeria

Abuja, Nigeria

Bachelor of Science (Hons) in Computer Science; 4.42/5.00 GPA

2024–2026

- **Relevant Coursework:** Operations Research, Numerical Analysis, Electronics Systems, System Design and Analysis.
- **Thesis:** Cognitive Vision-Based Hand Gesture Recognition System for Media and Accessibility Control.

University of Pécs

Pécs, Hungary

Bachelor of Science in Computer Science (Transferred); 4.24/5.00 GPA

2019–2023

- Software Engineering Concentration.

RESEARCH INTERESTS

Core Focus: Touchless Assistive Interfaces, Cognitive Vision for Frustration Detection, Age-Adapted Gesture Recognition Systems, Hardware Prototyping, and Signal Measurement.

RESEARCH EXPERIENCE

Cognitive Vision Hand Gesture Recognition System

Abuja, Nigeria

Undergraduate Thesis / Final Year Project

2025

- Developed a two-handed gesture system that controls system volume, brightness, and media playback on a consumer CPU. This was achieved by incorporating geometry mappings and focusing only on the necessary physical actions of hand landmarks mapped to digital actions to reduce the user's cognitive load. Average latency of 35 ms; ~28–30 FPS in good lighting, ~15–20 FPS in low light on consumer-grade CPUs.
- Implemented geometric regression algorithms, linear interpolation, and palm-depth proxies (Z-axis) alongside a gesture clutch mechanism to eliminate false activations and tremor-induced signal jitter.
- Selected to showcase the system at the faculty thesis exhibition panel.

PROJECTS

ROS Robot – Controller Input: Configured teleoperation nodes and subscriber/publisher communication pipelines for a chassis robot built on Ubuntu OS and ROS Noetic. Integrated Arduino microcontrollers. Tech: Raspberry Pi 3 Model B, Arduino, CamJam EduKit, ROS Noetic, Python.

Pathfinding Visualization GUI: Implemented Dijkstra's Algorithm, Breadth-First Search (BFS), and A* Search for graph transversal. All algorithms try to find the shortest path from one point to another.

Not3s – Cross-Platform Note-Taking Application: Developed a cross-platform note-taking application using the Flutter SDK. It uses Firebase Cloud Messaging `whisper_kit` to enable data synchronization and accurate on-device dictation.

PROFESSIONAL EXPERIENCE

Automation Developer Intern

Budapest, Hungary

ExxonMobil

2021–2022

- Worked closely with senior developers to design, develop, and showcase 3 different use-cases for Pipedream's integration layer, resulting in an increase of new workflows that adapted the cloud infrastructure over 3 months.
- Designed and implemented 100+ test cases using python for testing of complex in-house tooling systems and achieved 80% test coverage.
- Successfully worked in an agile development environment, contributing to the elimination of manual pipeline bottlenecks.

Software Developer Intern

Lagos, Nigeria

WebSoft Inc

2018–2019

- Migrated the entire callback microservices solution codebase from Python 2 to Python 3, consisting of 2,000 lines of code. Resolved compatibility issues and reduced external package dependencies by 25%.
- Designed and built RESTful API microservices hosted on Google Cloud Run serverless platform.
- Communicated with clients and senior project managers during sprint planning sessions.

Mid-Semester Summer Instructor

Pécs, Hungary

Summer Camp, University of Pécs

2019–2023

- Provided academic support in VBA to automate real-world data workflows to mostly other freshmen.
- Graded assignment, exams, and discussion sections, while providing individualized feedback to over 20+ students.

SKILLS

Programming: Python, Dart, C++.

Technical: Firebase, MySQL, OpenCV, TensorFlow, ROS, Arduino, GPIO Interfacing.

Version Control & Tools: Git/GitHub, Linux, $\text{\LaTeX}$.

Languages: English (Native), Hungarian (Elementary).

AWARDS

Dean's Merit Award

University of Pécs

Pécs, Hungary
2021–2022 & 2022–2023